

Course Description

Module Title

English title: Introduction to Data Handling, Exploration & Applied Machine Learning

Danish title: Introduktion til datahåndtering, udforskning og anvendt maskinlæring

ECTS

Number of ECTS: 10

Location

Education programme: Master of Science in Economics and Business Administration (Business Data Science)

Semester: 1st

Module Coordinator

Module coordinators:
Roman Jurowetzki
Hamid Bekamiri

Type and Language

Module type: Course

Language of instruction: English

Learning Objectives

KNOWLEDGE

The objective is that the student after the module possesses the necessary knowledge on:

- the main workflow routines and techniques how to obtain, store, manipulate, and analyse data using relevant software and foundational machine learning approaches.
- how to use a wide variety of techniques to explore, visualise, describe and present their data, including unsupervised and supervised machine learning techniques.
- the differences and complementarities between the prediction focussed data science approach, and the causality seeking approach of traditional scientific statistics.

SKILLS

The objective is that the student after the module possesses the necessary skills in:

- reading, importing, exporting, manipulating, cleaning and (pre)processing data in most widely used data formats using relevant software.
- selecting and applying standard techniques from 'traditional' statistics and data science to solve empirical problems of data exploration, classification, optimisation, and forecasting including model performance evaluation, fine-tuning and optimisation.
- understanding, interpreting, critically reflecting upon, and explaining the results of data analysis.

COMPETENCES

The objective is that the student after the module possesses the necessary competences in:

- comprehending and participating in current professional and academic discussions in applied statistics and data science as well as autonomously expanding their knowledge in the field.
- critically reflecting possibilities and constraints related to the implementation and evolution of data-driven methods, including Identify problems which can be wholly or partially solved by the use of data analytics.
- describing and communicating the results of data analysis in a precise, understandable and informative manner, using appropriate data description and visualisation techniques.

Academic Content and Conjunction With Other Modules/Semesters

This module intends to provide an opportunity to sample the core techniques of data science and machine learning, understand their intuition and application cases. It also aims at showing best practice of how to select specific and appropriate methods for the particular data science project, as well as how to efficiently and autonomously acquire further knowledge of the rapidly evolving field. Insights and techniques learned in this module can be applied to real-world problems in, e.g., marketing (How do you classify customers who are likely to spend a lot?), management (How do you identify performance bottlenecks in the organisation?) or finance (Is this person likely to default on their mortgage?).

Upon completion of the module students will have built a solid and expandable knowledge foundation in modern data science and will have acquired a broad range of skills enabling them to carry out own data analysis projects. More specifically the module will cover foundations of data manipulation, exploratory data analysis, supervised and unsupervised machine learning. Students will be capable of autonomously managing and evaluating complex projects and problems associated with data management, description, and analysis.

Scope and Expected Performance

10 ECTS corresponds to a workload of 270 hours, distributed among the following activities:

With teacher (including guest teachers and TAs):

Lectures: 40 hours

Exercises/Seminar: 20hours

In total: 60 hours

Without teacher:

Exercises/Casework: 80 hours

Increased focus on assignment preparation and submission, including in-depth casework and exercises.

Curriculum preparation: 130 hours

Additional time allocated for independent study, reading, and deep understanding of the curriculum.

In total: 210 hours

Module Activities (Course Sessions etc.)

1. Session: Introduction to Data Science: Tools and Foundations

- **Description:** The opening session establishes the module's foundation: an introduction to the faculty team, the Python data-science ecosystem (Jupyter notebooks, scientific libraries), version control with Git/GitHub, and shared cloud-computing resources for the module. The session also introduces the module's working approach to AI coding assistants — what they are useful for, how to use them productively, and the cases in which the module asks students to work without them.
- **Recommended Readings:**
 - VanderPlas, J. (2016). *Python Data Science Handbook: Essential Tools for Working with Data*. O'Reilly Media, Inc.

2. Session: Data Handling and Manipulation with Pandas & NumPy

- **Description:** The core techniques for working with structured data: students learn to use Pandas and NumPy to load, clean, filter, group, aggregate, join and reshape datasets of realistic size and messiness. The skills established here underpin every subsequent session and are practised throughout the module's weekly portfolio assignments.
- **Recommended Readings:**
 - VanderPlas, J. (2016). *Python Data Science Handbook: Essential Tools for Working with Data*. O'Reilly Media, Inc.

3. Session: Exploratory Data Analysis and Applied Statistics

- **Description:** Students learn to investigate datasets systematically — uncovering patterns, identifying anomalies, testing hypotheses, formulating new questions — and to apply the statistical concepts that underpin valid interpretation of the results. The session covers the choice and execution of common tests (t-tests, ANOVA, Chi-Square), distinguishes statistical significance from business significance, and works through a realistic case study on workforce analytics.
- **Recommended Readings:**
 - Nield, T. (2022). *Essential Math for Data Science*. O'Reilly Media.

4. Session: Communicating Insights with Data Visualization

- **Description:** The principles of effective data visualisation: choosing the right chart for a question, designing for clarity, building interactive views. Students gain hands-on experience with the main Python visualisation libraries (Matplotlib, Seaborn, Altair) and learn to construct visual narratives that make analytical findings legible to non-technical stakeholders.
- **Recommended Readings:**
 - Wilke, C. O. (2019). *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures*. O'Reilly Media.

5. Session: Rapid Prototyping with Streamlit, Part I: Building Applications

- **Description:** Students learn to wrap analyses into interactive, browser-based applications using Streamlit, the leading Python framework for analytical app development. The session covers UI

construction, data display, state management and the conversion of static analyses into shareable tools that non-technical stakeholders can use.

- **Recommended Readings:**

- Khorasani, M., Abdou, M., & Hernández Fernández, J. (2022). *Web Application Development with Streamlit*. Apress (selected chapters).

6. Session: Rapid Prototyping with Streamlit, Part II: Deployment & Best Practices

- **Description:** Continuing from Part I, students learn to optimise, structure and deploy interactive applications to cloud environments. Topics include performance caching, efficient columnar storage and query engines for handling larger datasets, environment and secrets management, and the deployment workflow itself.
- **Recommended Readings:**
 - Khorasani et al. (2022), continued.

7. Session: Unsupervised Learning I: Dimensionality Reduction

- **Description:** The techniques for finding the “essential few” variables that explain complex patterns: Principal Component Analysis and related linear methods, and modern visualisation approaches such as UMAP and t-SNE. Students apply these to a multi-attribute dataset (e.g., factors that make cities attractive to remote workers) to discover structure in high-dimensional data.

8. Session: Unsupervised Learning II: Clustering Analysis

- **Description:** Algorithms for identifying natural groups in data — K-means, hierarchical and density-based clustering — and the diagnostics that distinguish meaningful structure from noise. Students work through different clustering cases and learn how feature engineering and scaling decisions silently shape cluster outputs.

9. Session: Recommendation Systems and Similarity Search

- **Description:** The mathematics and practice of recommender systems: collaborative filtering, similarity-based recommendation, and the use of dimensionality reduction (e.g., TruncatedSVD) to make similarity search efficient. Students build a working recommender on a real dataset and deploy it as a small interactive app. The session closes with a discussion of how modern LLMs are reshaping recommendation systems in industry.
- **Recommended Readings:**
 - Leskovec, J., Rajaraman, A., & Ullman, J. D. *Mining of Massive Datasets*, 3rd ed. — Chapter 9: Recommendation Systems. Plus a current survey on LLM-enhanced recommender systems.

10. Session: Workflow Automation in Data Science

- **Description:** A bridging session connecting the unsupervised techniques of the first half of the module to the supervised pipeline of the second. Students learn the practical construction of data and analytical workflows using industry-standard automation platforms (e.g., n8n), integrating language-model APIs where useful, and — critically — building the small evaluation harnesses that determine whether an automated workflow meets its business requirements. The same evaluation-harness discipline is then carried forward into the supervised machine-learning sessions.
- **Recommended Readings:**
 - Ng, K. K. H., et al. (2021). A systematic literature review on intelligent automation.

11. Session: Introduction to Supervised Machine Learning

- **Description:** The fundamentals of supervised learning: regression and classification, train–test splitting, evaluation metrics (RMSE, precision, and recall), the bias–variance tradeoff, and model interpretation with SHAP values. Students walk through a complete pipeline from raw data to interpreted model on a realistic business case.

12. Session: Advanced Supervised Learning: Ensembles and Optimization

- **Description:** Students extend the supervised-learning workflow to gradient-boosted ensembles (XGBoost) and address the practical concerns that turn a notebook into a production system: temporal train–test splits to prevent data leakage, model calibration, cost-based decision thresholds aligned with business cost matrices, and end-to-end deployment of a trained model as an interactive web service (FastAPI backend, Streamlit user interface). The framing is decision-theoretic throughout: model outputs are evaluated against quantifiable business costs, not accuracy alone.

13. Session: Advanced Topics: Time Series Analysis

- **Description:** Methods for forecasting and anomaly detection over temporal data: decomposition into trend, seasonality and irregular components, time-aware train–test splits, feature-rich forecasting with engineered lags and exogenous variables, evaluation with MAE / RMSE / MAPE, and anomaly detection from model residuals and prediction intervals.

14. Session: Module Assignment Workshop

- **Description:** A dedicated session to kick off the final module assignment, providing guidance on project scope, requirements, and best practices for success.

15. Session: Final Q&A and Assignment Support

Description: An open forum for questions and answers related to the module assignment, providing students with a final opportunity for clarification and support before submission.

Resources

Core Textbooks:

- VanderPlas, J. (2016). *Python Data Science Handbook: Essential Tools for Working with Data*. O'Reilly Media, Inc.
- Wilke, C. O. (2019). *Fundamentals of Data Visualization: A Primer on Making Informative and Compelling Figures*. O'Reilly Media.
- Nield, T. (2022). *Essential Math for Data Science*. O'Reilly Media.

Additional Resources:

- Official documentation for Python libraries and packages used throughout the course (e.g., Pandas, NumPy, Scikit-learn, Streamlit).
- All presentation slides and supplementary materials will be available on Moodle.

Theory and Industry Context:

A curated collection of 5-7 current articles and case studies will be provided to illustrate the application of

AI, Machine Learning, and Data Science in real-world business scenarios.

Examination

Name of exam: Introduction to Data Handling, Exploration & Applied Machine Learning

Type of exam: Oral exam

Individual examination.

Assessment: 7-point grading scale

Type of grading: Internal examination

Prerequisite for participation in the exam:

- A prerequisite for participating in the exam is that the student has handed in written material.

How is the exam conducted:

The written material submitted by the student prior to the exam consists of a portfolio developed across the module: a series of weekly assignments practising the techniques covered in each session; short group-produced video presentations in which students walk through and discuss their solutions (uploaded as recordings of group screen-shared sessions); and a final assignment submitted in the last week of the module, which is the central artefact discussed at the oral examination.

Students receive a question or problem to prepare 30 minutes before the oral examination, with access to specified preparation materials during this period. The examination begins with the student's presentation of the prepared question, followed by a Q&A session covering the course curriculum and the submitted portfolio material. During the examination, the student has access to the materials used during the 30-minute preparation period (scope and amount specified with the question) and to their submitted portfolio.

Allowed aids: A computer with internet access may be used to demonstrate non-written elements of the project assignment (e.g., applications, computational notebooks, code). Otherwise, only a printed or on-screen PDF version of the project is permitted. Communication with others and the use of generative AI tools during the examination are not allowed, except when demonstrating functionality that was developed as part of the project.

Information about the re-exam:

According to the examination regulations, section 1.1 on the determination of time and place for holding exams, you as students can expect that information about the exam form, time, place, and permitted aids will be communicated no later than 2 weeks before the re-exam is held. If the re-exam is conducted digitally, this will also be communicated no later than 2 weeks before the re-exam is held.

For exam date and further information go to Moodle.

